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$$\begin{aligned}\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} &= \frac{0}{0} \\ \stackrel{(H)}{=} \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} &= \frac{0}{0} \\ \stackrel{(H)}{=} \lim_{x \rightarrow 0} \frac{\sin x}{6x} \\ &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1}{6} = \frac{1}{6} \\ \text{want } \lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1\end{aligned}$$

References